

# 高质量美赛精美绘图实战解析课

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- 1 论文流程图制作及个人心得
- 2 常用实验结果可视化方法
- 3 机器学习画图模板
- 4 部分可视化工具及组件展示

# 1

## 论文流程图制作及个人心得

# 团队论文流程图展示1

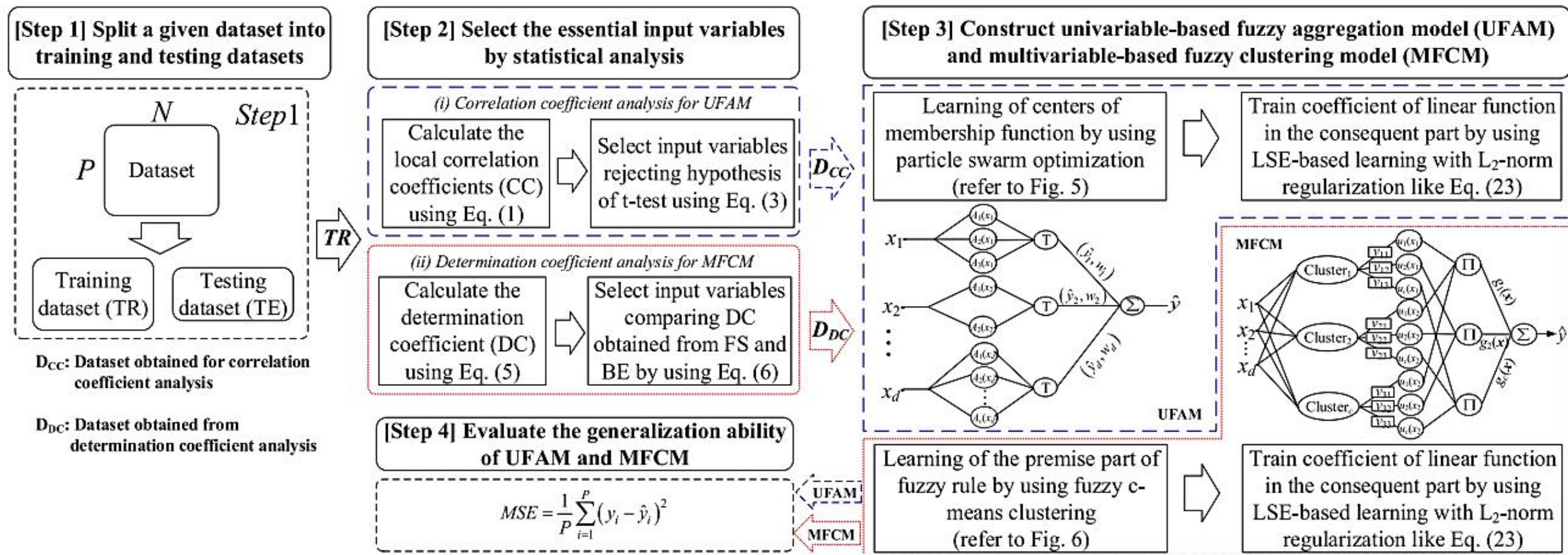
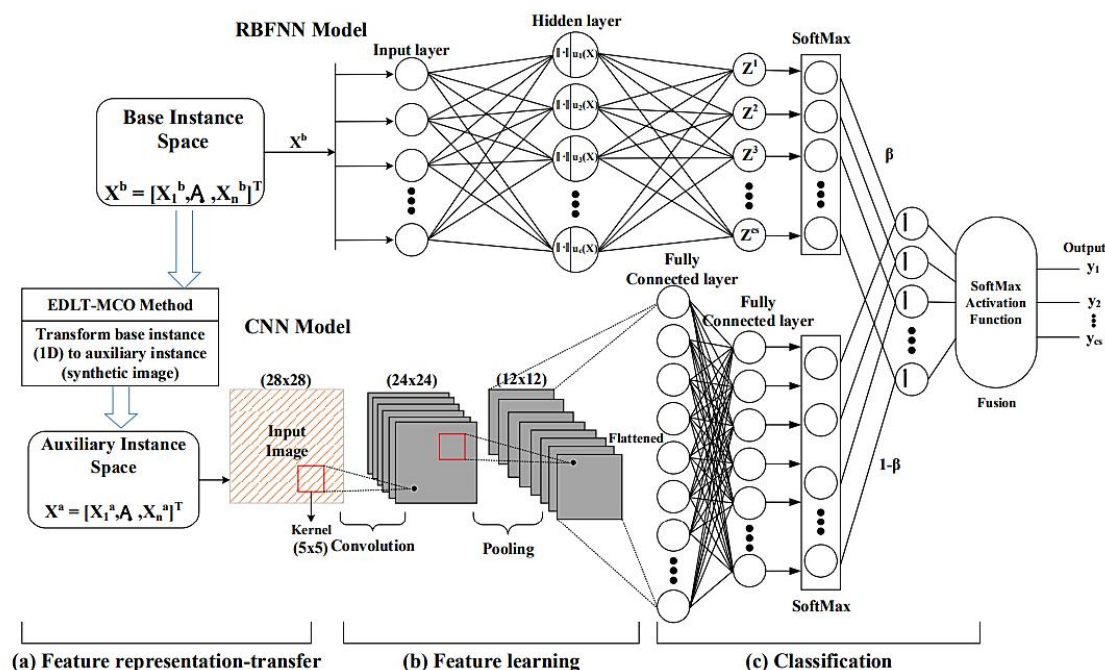
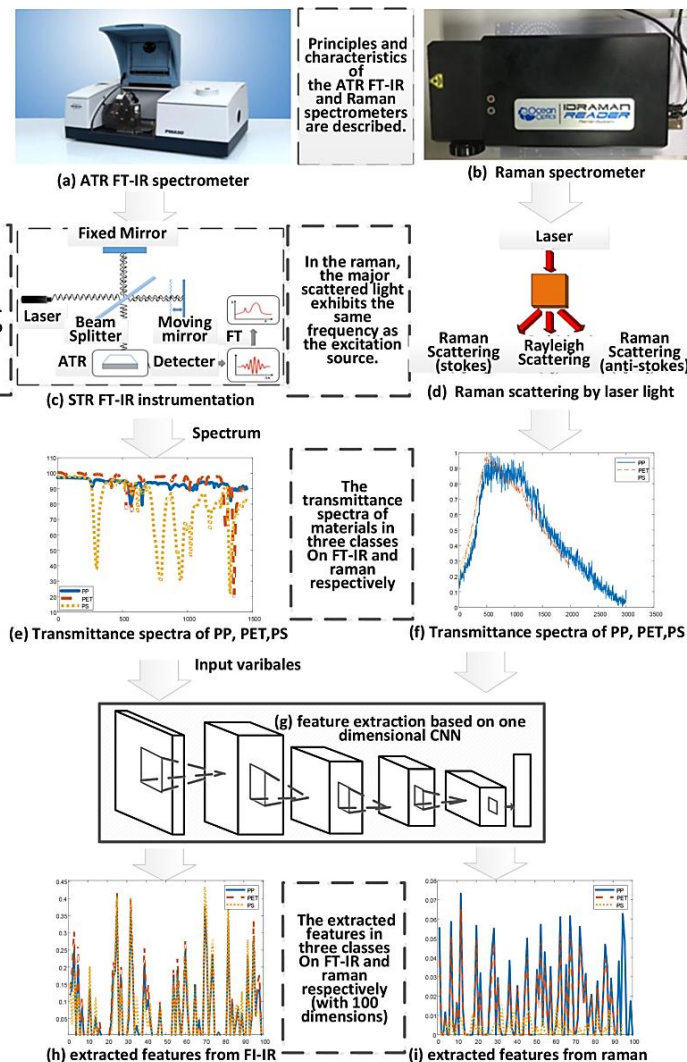


图1 基于单变量相关和多变量确定系数分析的特征选择强化模糊模型设计

# 团队论文流程图展示2



- 将文字、流程还有实验结果图相结合，使图内容更丰富；
- 注意结构编排，从上到下或者从左到右，过于混乱的结构会影响读者理解；
- 注意颜色的运用，要具备基本色彩美感；

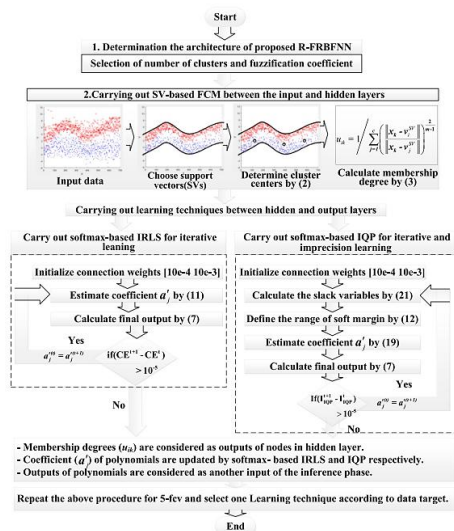
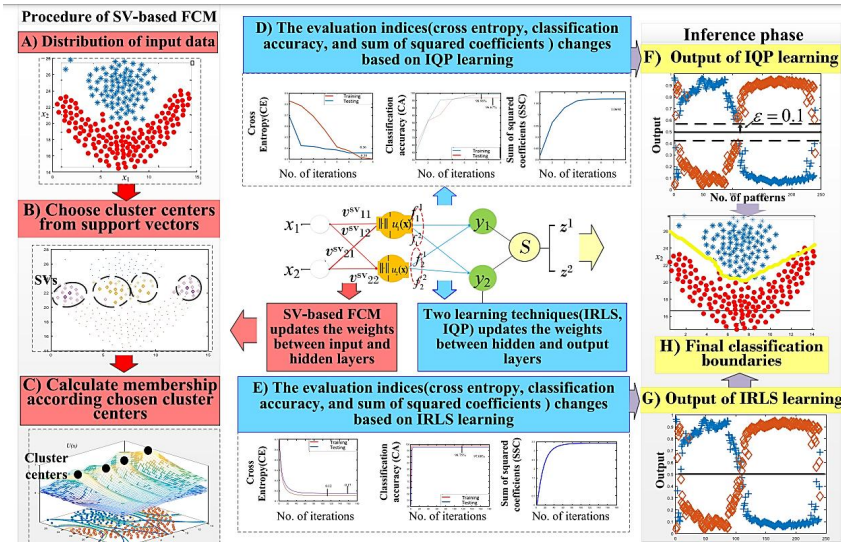
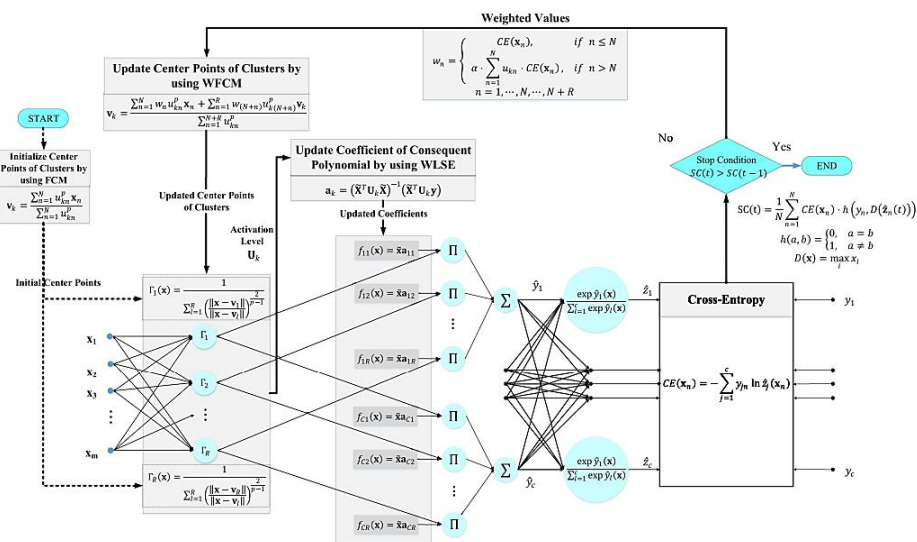
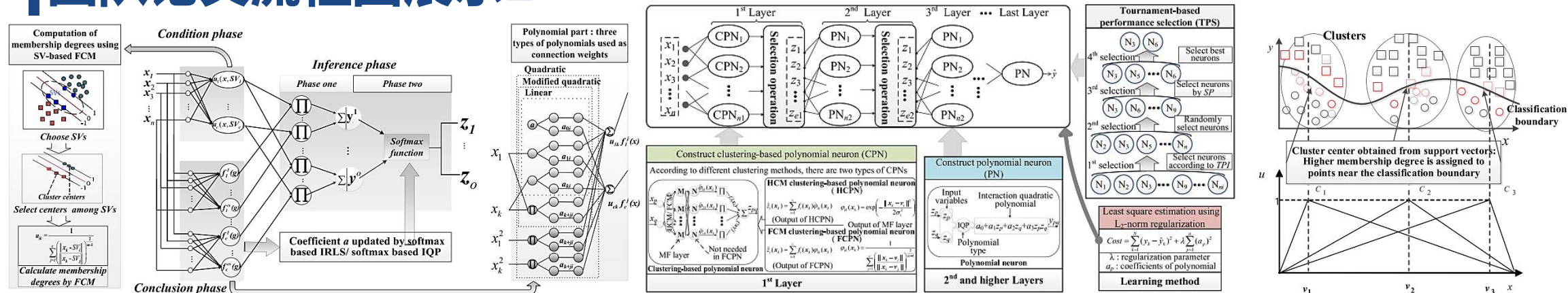


[2] Yang, C., Oh, S. K., Yang, B., Pedrycz, W., & Wang, L. (2022). Hybrid fuzzy multiple SVM classifier through feature fusion based on convolution neural networks and its practical applications. *Expert Systems with Applications*, 202, 117392. (IF:8.665)

[3] Zhou, K., Oh, S. K., Qiu, J., Pedrycz, W., & Seo, K. (2022). Reinforced Two-stream Fuzzy Neural Networks Architecture Realized with the Aid of 1D/2D Data Features. *IEEE Transactions on Fuzzy Systems*. (IF:12.253)



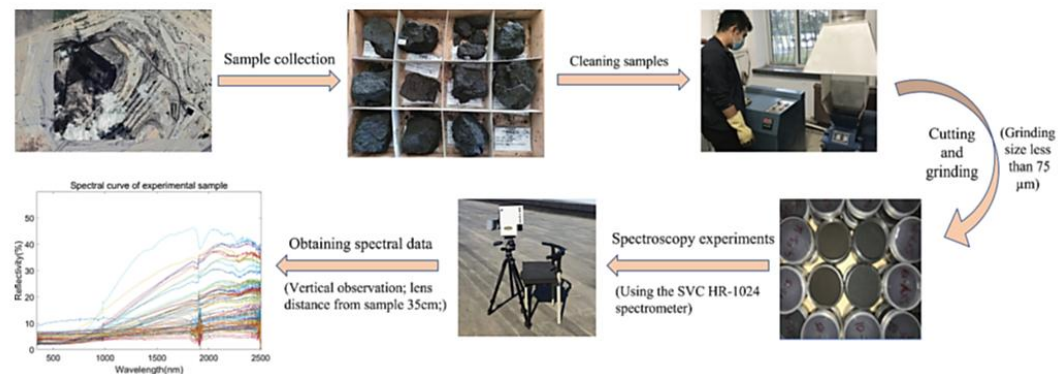
## 团队论文流程图展示2



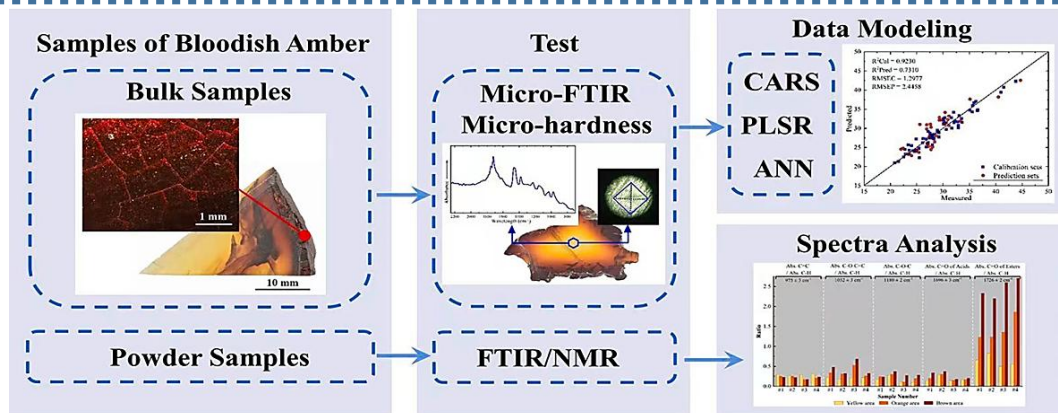
- [4] Yang, C., Oh, S. K., Yang, B., Pedrycz, W., & Wang, L. (2022). Hybrid fuzzy multiple SVM classifier through feature fusion based on convolution neural networks and its practical applications. *Expert Systems with Applications*, 202, 117392. (IF:8.665)
- [5] Zhou, K., Oh, S. K., Qiu, J., Pedrycz, W., & Seo, K. (2022). Reinforced Two-stream Fuzzy Neural Networks Architecture Realized with the Aid of 1D/2D Data Features. *IEEE Transactions on Fuzzy Systems*. (IF:12.253)
- [6] Yang, C., Oh, S. K., Pedrycz, W., Fu, Z., & Yang, B. (2020). Design of Reinforced Fuzzy Radial Basis Function Neural Network Classifier Driven With the Aid of Iterative Learning Techniques and Support Vector-Based Clustering. *IEEE Transactions on Fuzzy Systems*, 29(9), 2506-2520. (IF:12.253)



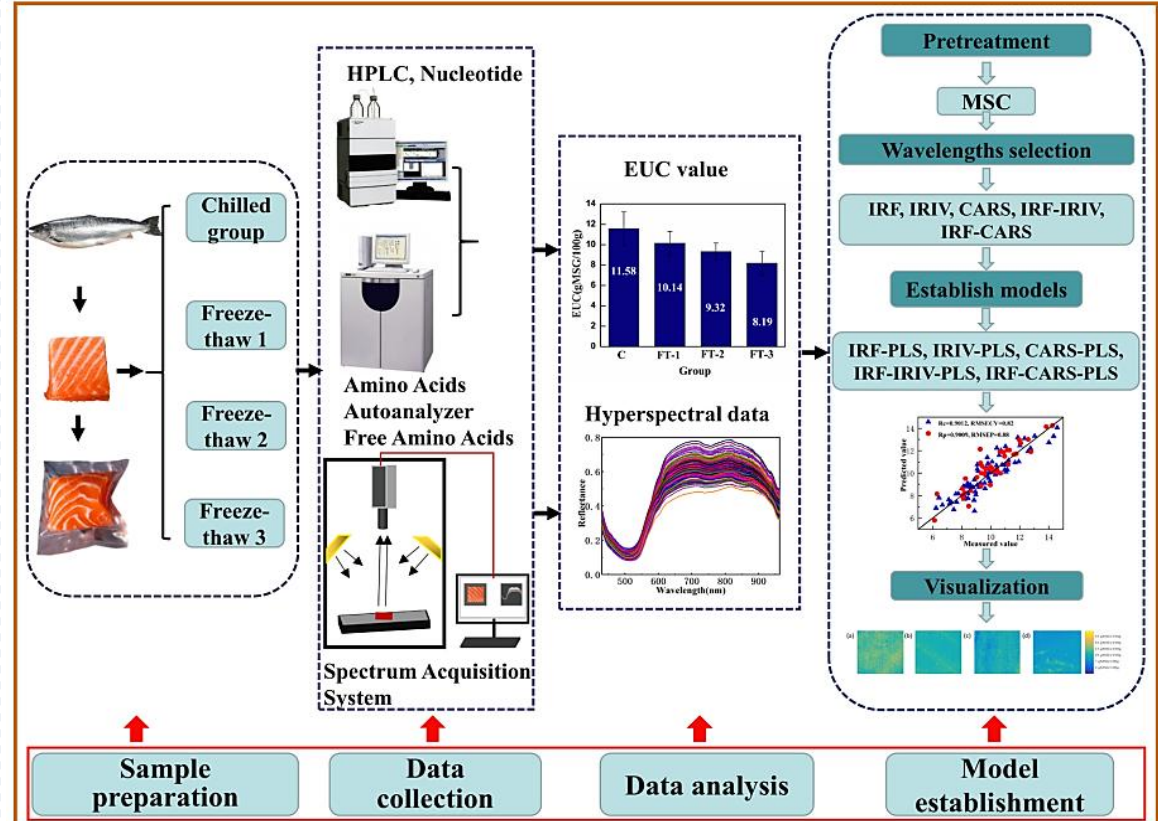
# 光谱论文流程图展示 (东北大学、江苏大学、中国地质大学)



基于反射光谱和深度学习的煤炭快速工业分析



利用PLSR和ANN模型的FTIR光谱评价缅甸琥珀的自然老化响应



使用高光谱成像技术结合多元算法对三文鱼中的等效鲜味浓度进行无损评估

- [7] Li, X., Wang, Y., Shi, G., Lu, R., & Li, Y. (2023). Evaluation of natural ageing responses on Burmese amber durability by FTIR spectroscopy with PLSR and ANN models. *Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy*, 285, 121936. (IF:4.831)
- [8] Sun, Z., Pan, H., Zuo, M., Li, J., Liang, L., Ho, C. T., & Zou, X. (2023). Non-destructive assessment of equivalent umami concentrations in salmon using hyperspectral imaging technology combined with multivariate algorithms. *Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy*, 285, 121890. (IF:4.831)
- [9] Xiao, D., Yan, Z., Li, J., Fu, Y., & Li, Z. (2023). Rapid proximate analysis of coal based on reflectance spectroscopy and deep learning. *Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy*, 287, 122042. (IF:4.831)



## 常用实验结果可视化方法

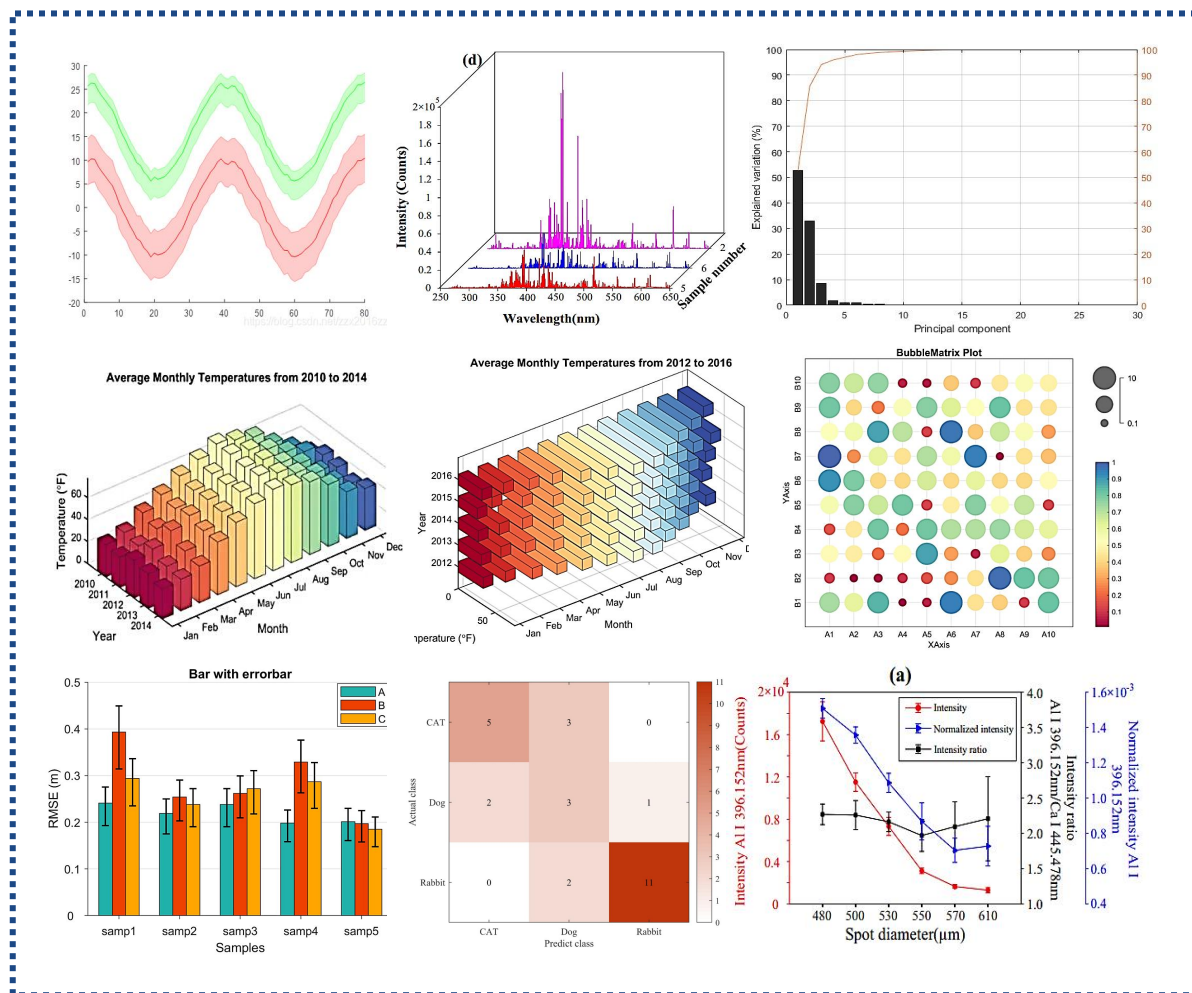




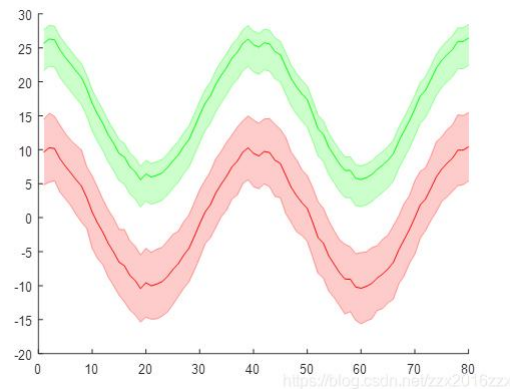
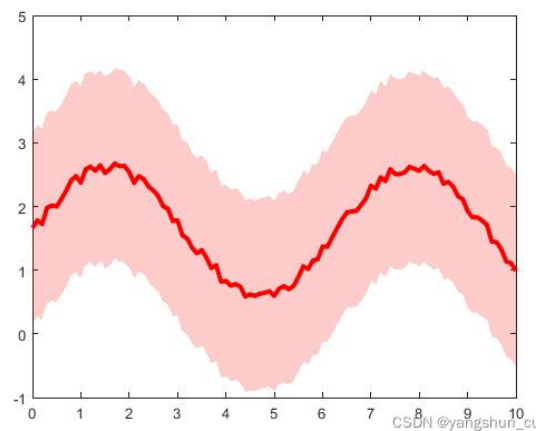
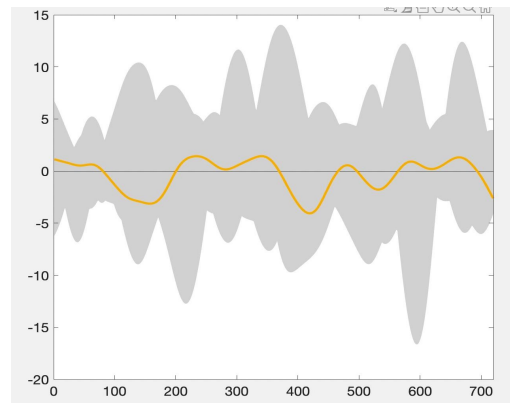
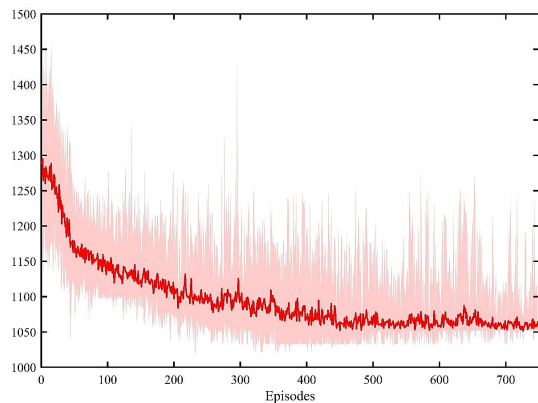


# 常用实验结果可视化图列表

- 带状曲线图
- 瀑布图
- 帕累托图
- 三维柱状图
- 水平三维柱状图
- 气泡矩阵图
- 带有误差棒的箱线图
- 混淆矩阵图
- 带误差棒的折线图



## 带状曲线图（带状误差线图）

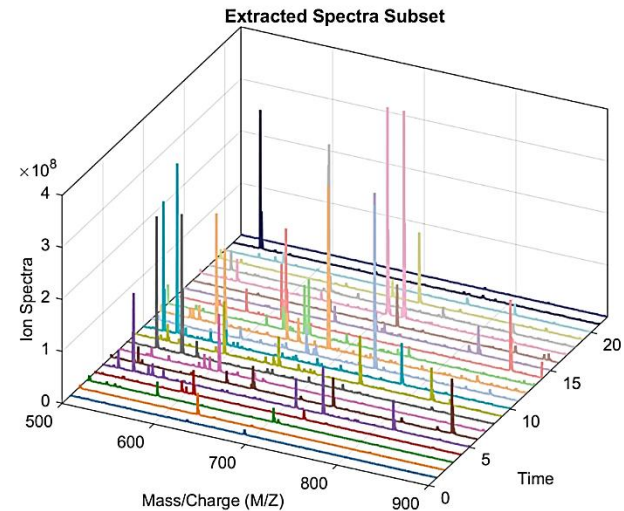
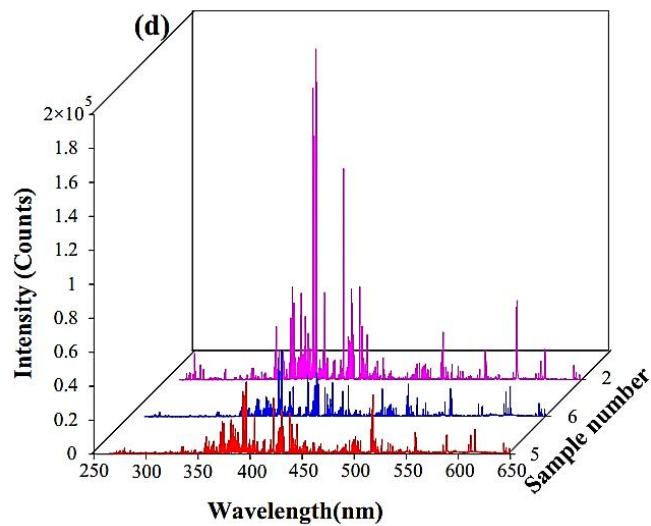
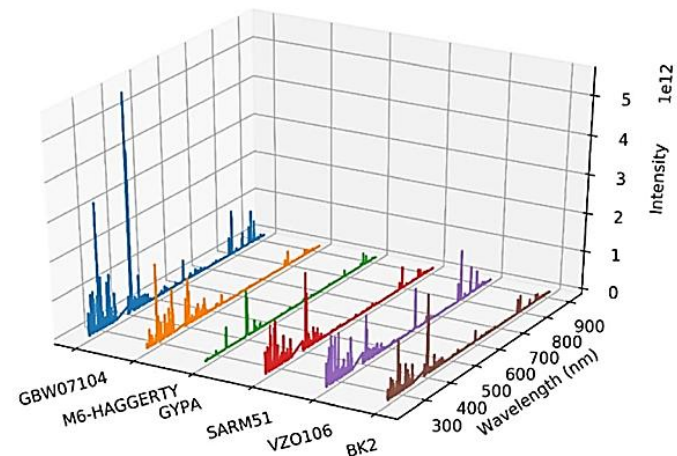
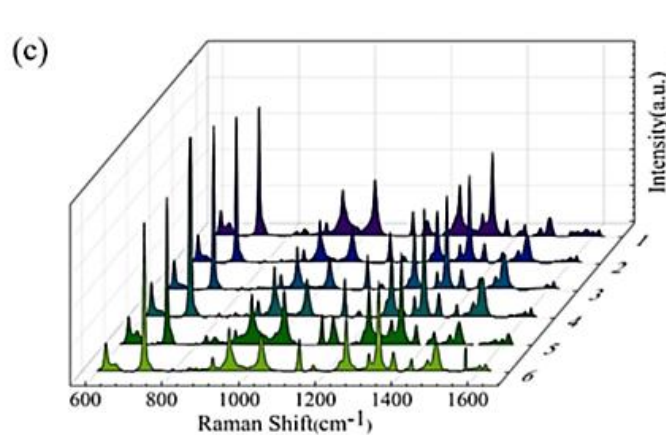


思路：将阴影当作一个由多段线连接而成的封闭区域，然后就该区域进行填充颜色；

MATLAB：‘patch’ 函数、下载 ‘shadedErrorBar’ 函数、‘area’ 函数



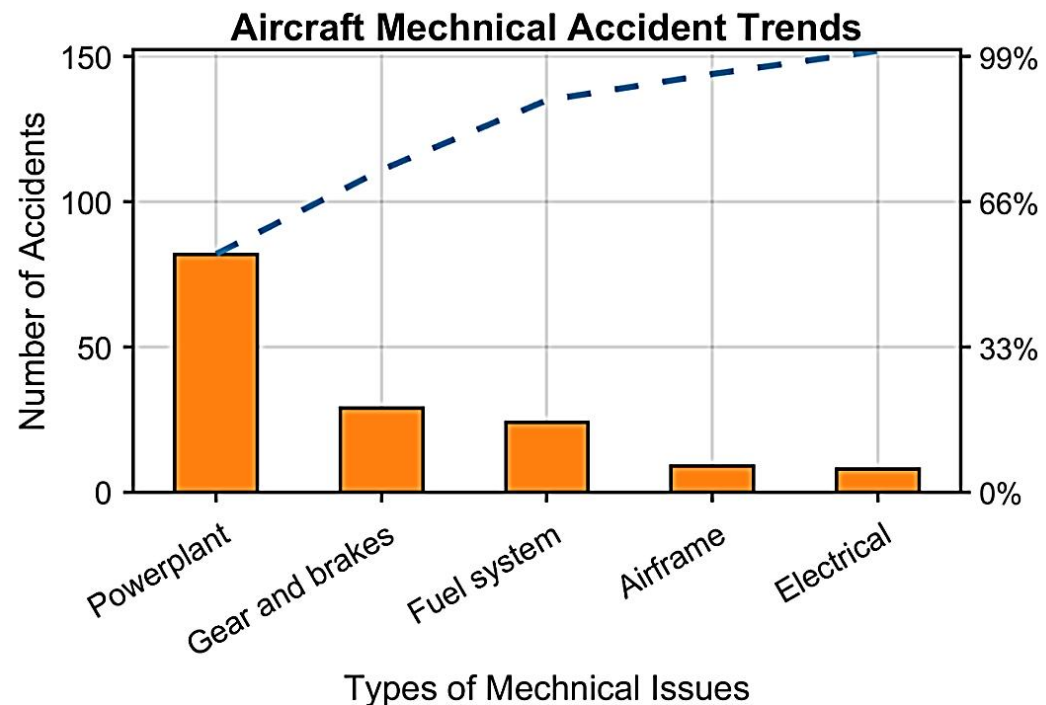
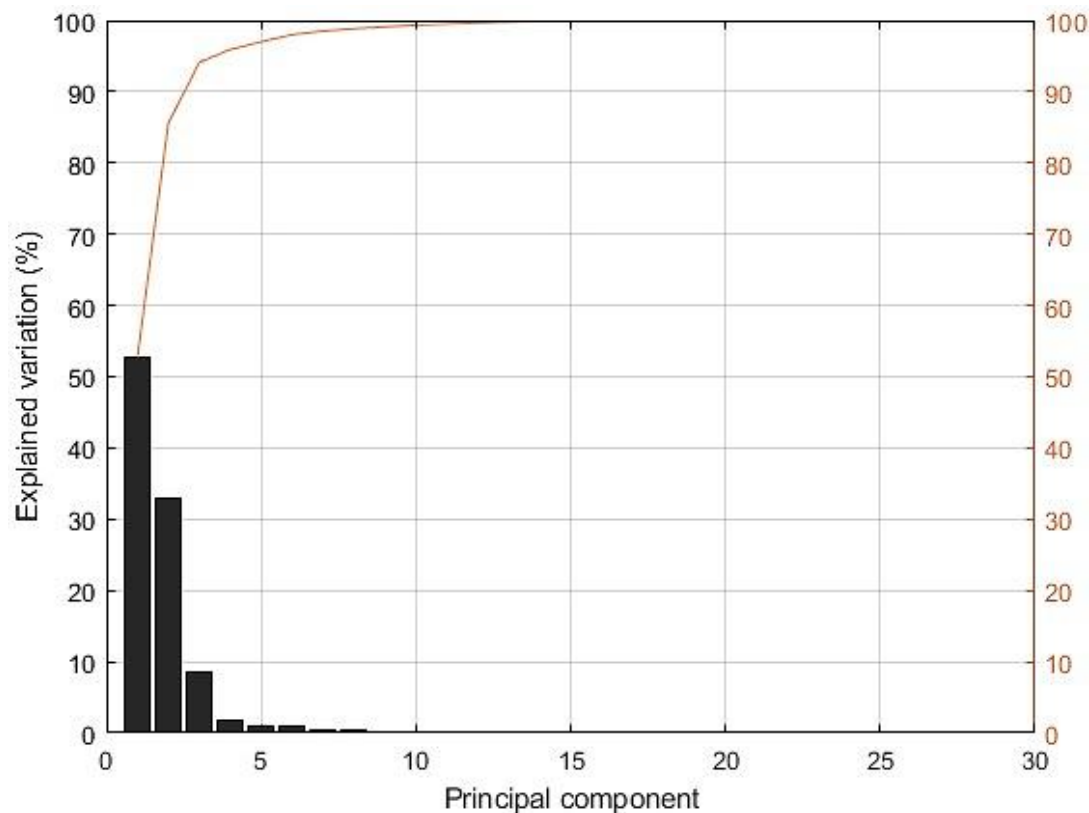
# 瀑布图



直接使用MATLAB中的plot3函数即可绘制



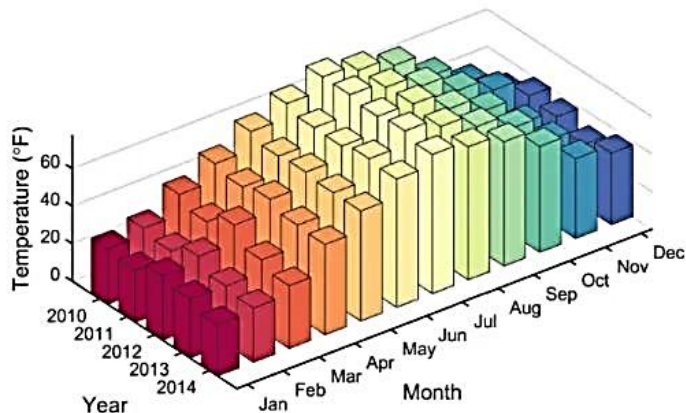
# 主成分分析——崖低碎石图（帕累托图）



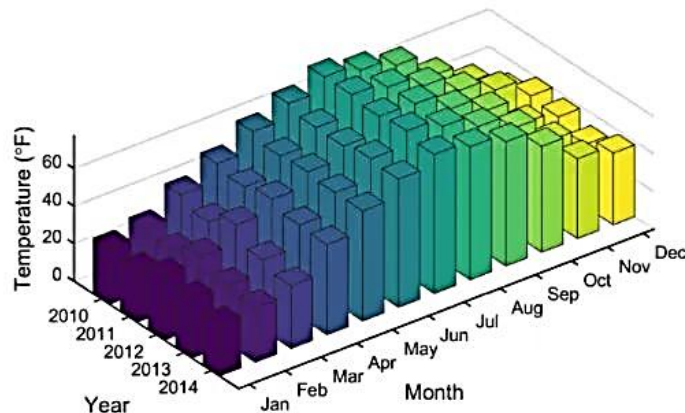
- 1.利用PCA获取每个主成分的方差（条形图）
- 2.求解方差累计贡献（折线图）
- 3.双y轴可做可不做

# 三维柱状图

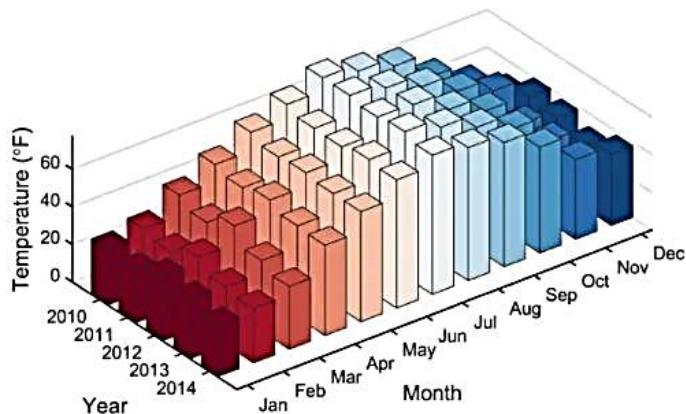
Average Monthly Temperatures from 2010 to 2014



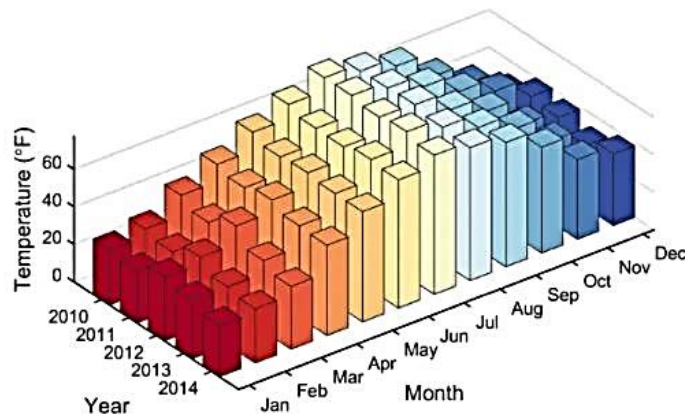
Average Monthly Temperatures from 2010 to 2014



Average Monthly Temperatures from 2010 to 2014

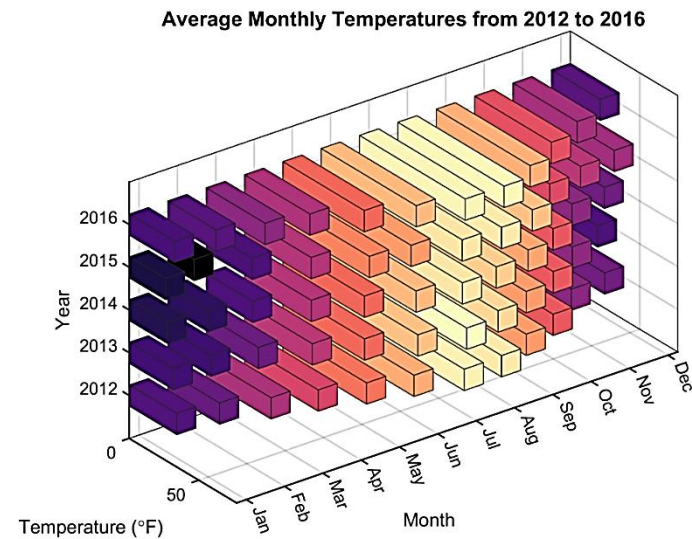
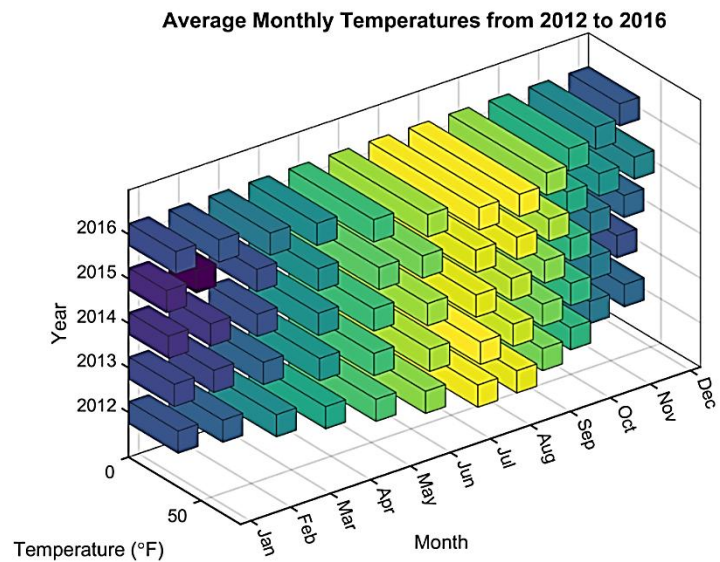
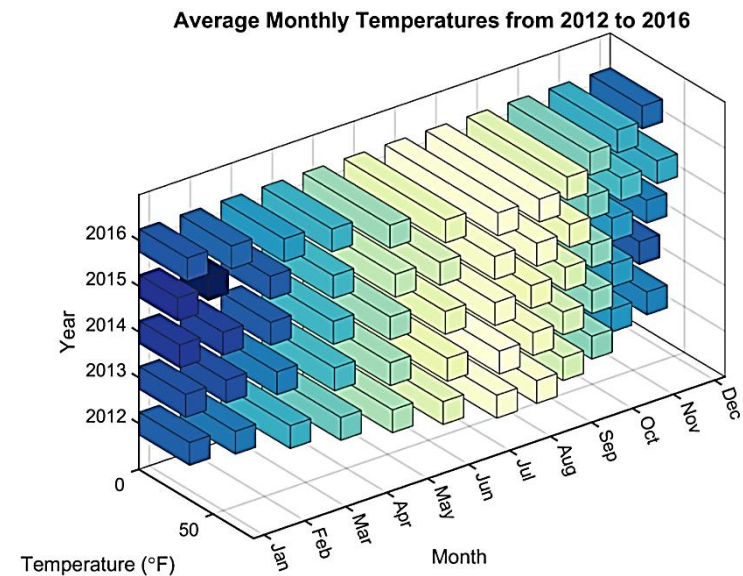
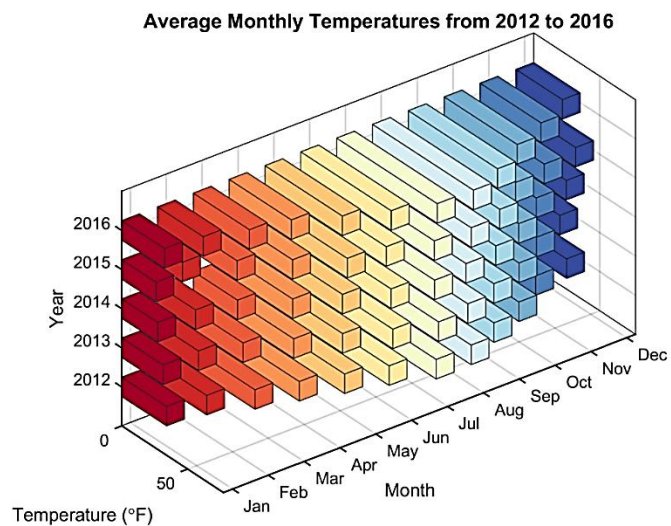


Average Monthly Temperatures from 2010 to 2014



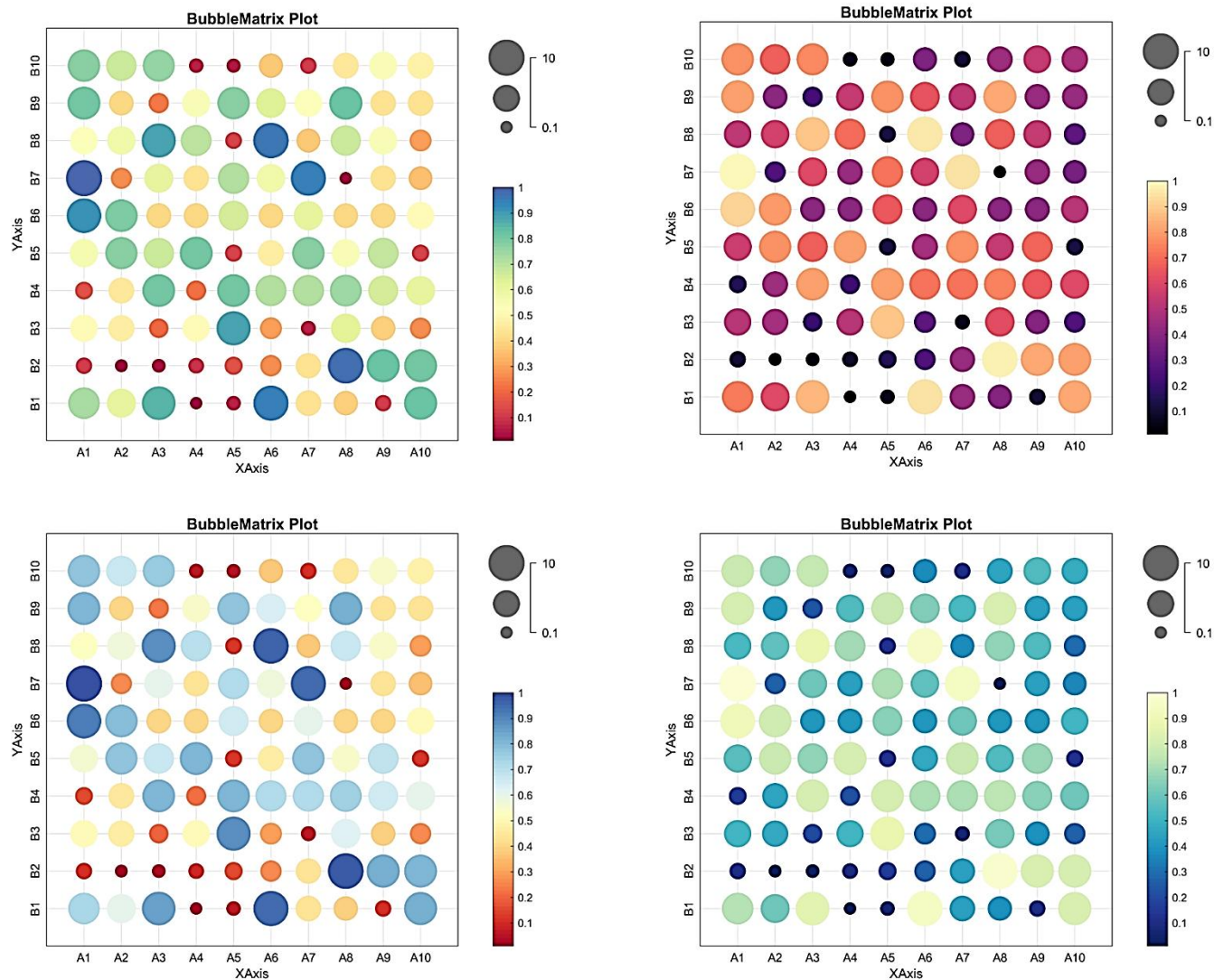
```
load BostonTemp.mat
yearIdx = 16;
TempData6Years
=Temperatures(yearIdx:yearI
dx+5,:);
%% 颜色定义
map = addcolorplus(300);
% map = flipud(map);
figureHandle = figure;
b =
bar3(TempData6Years,0.5);
hTitle = title(sprintf('Average
MonthlyTemperatures from
%d to
%d',Year(yearIdx),Year(yearI
dx+5)));
hXLabel = xlabel('Month');
hYLabel = ylabel('Year');
hZLabel =
zlabel('Temperature (\circ F)');
```

# 水平三维柱状图





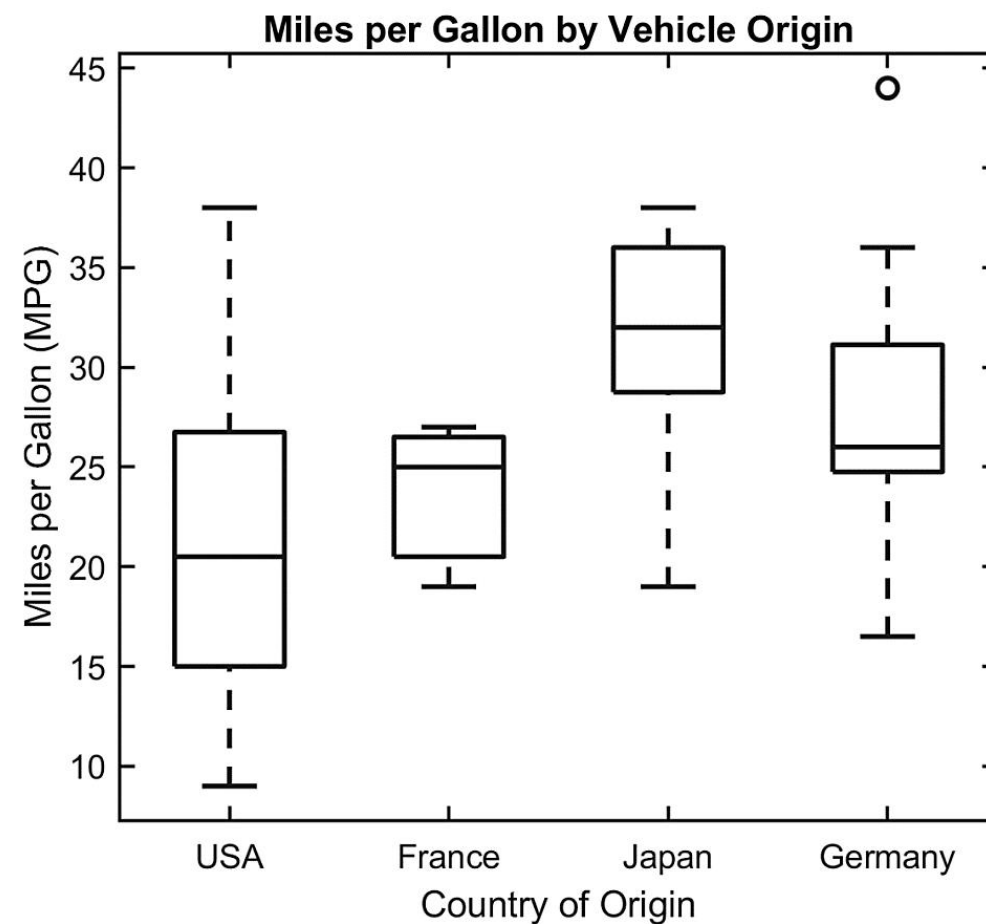
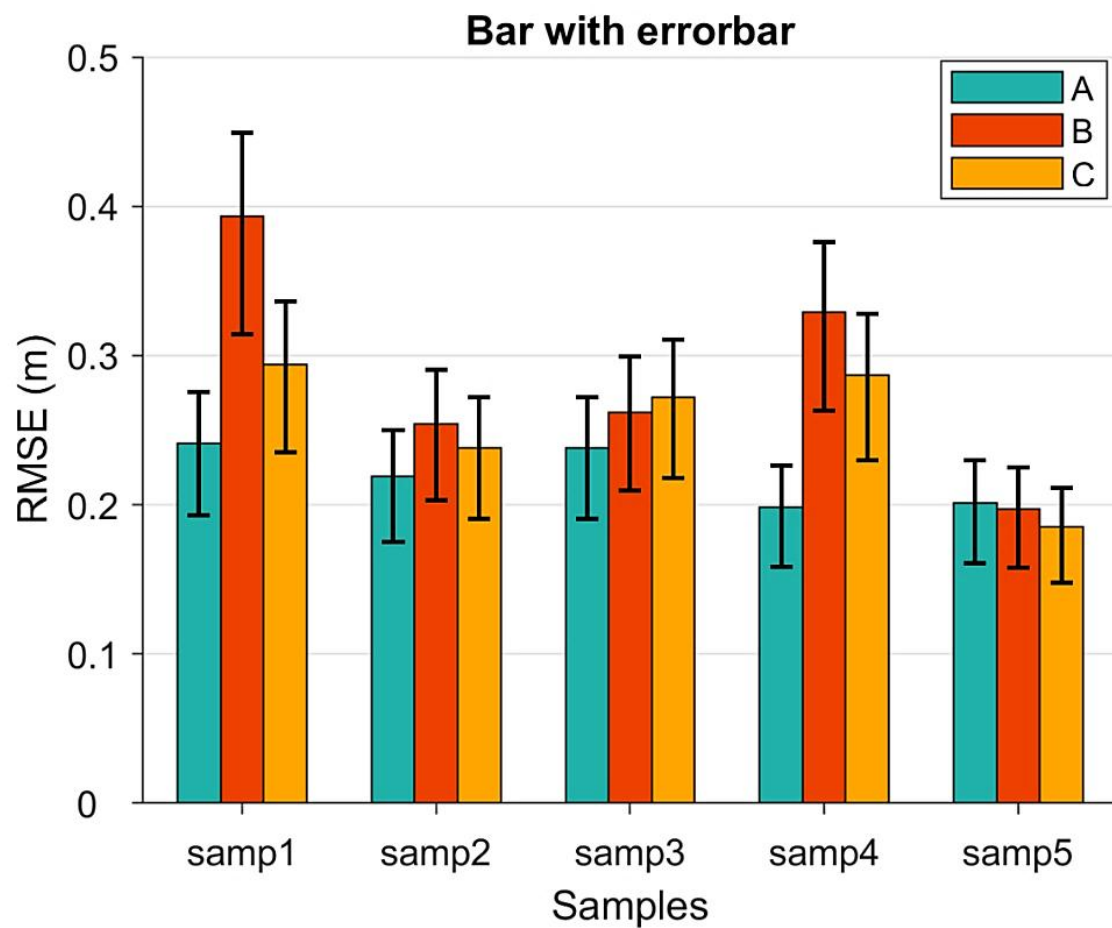
# 相关性矩阵图——气泡矩阵图



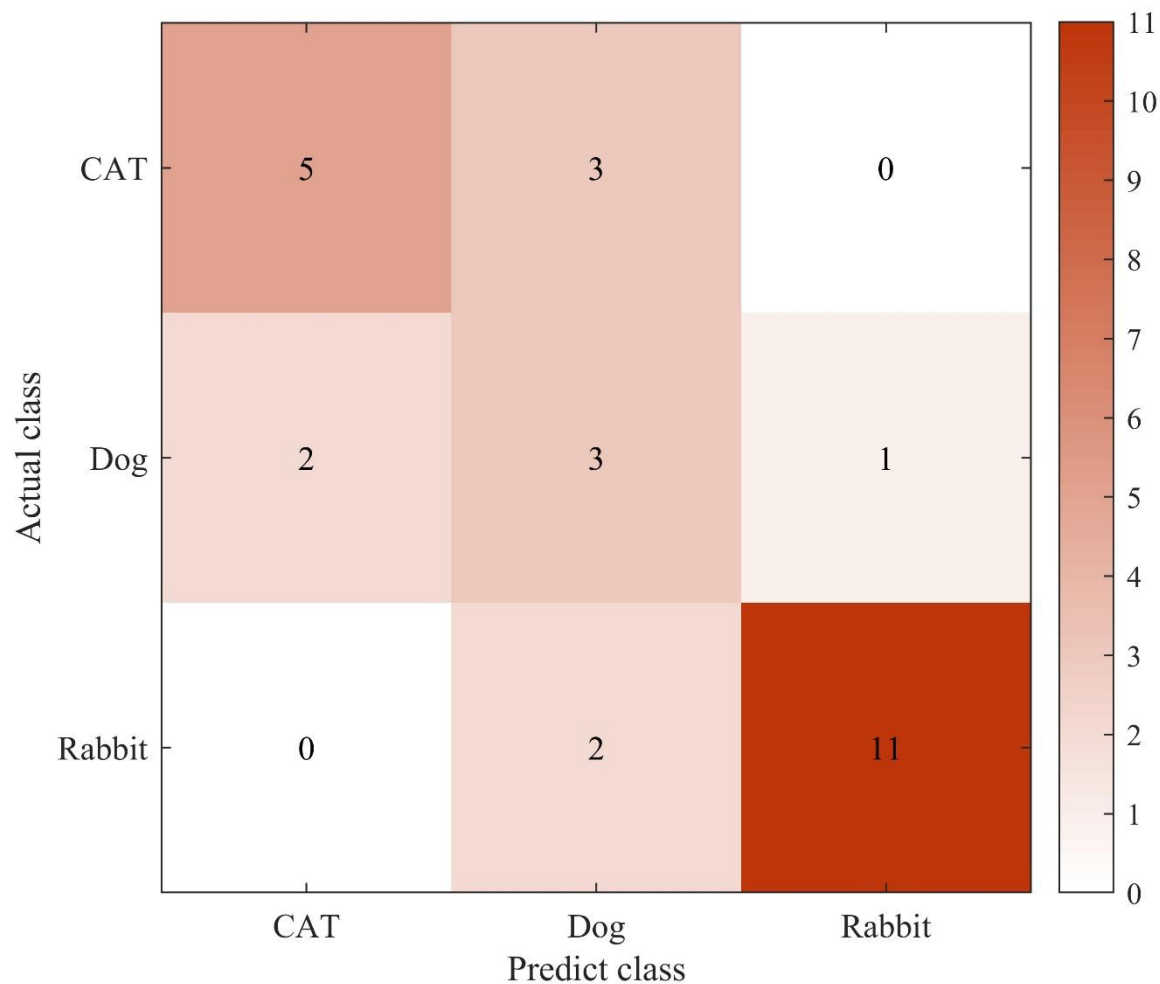
由于空间有限  
配套代码搜索：阿昆的科研日常



## 箱线图（带有误差棒）

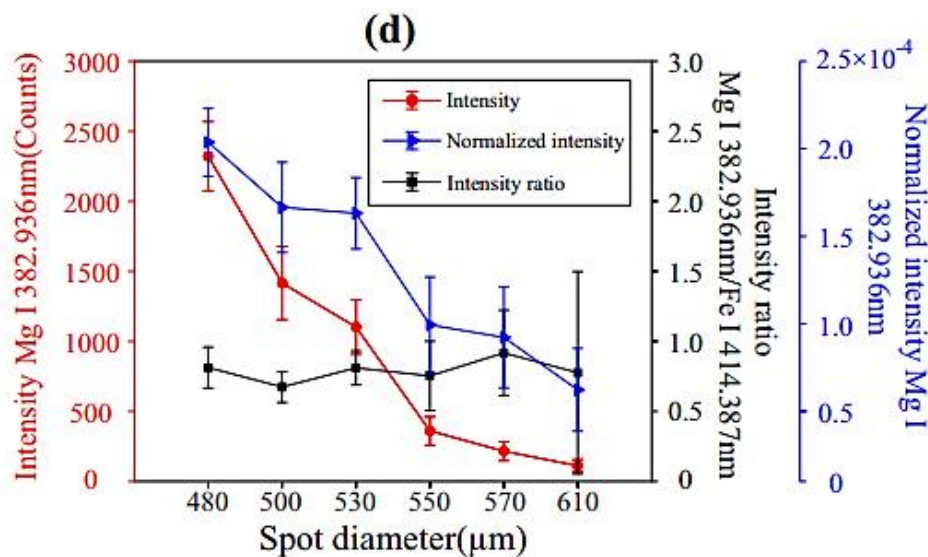
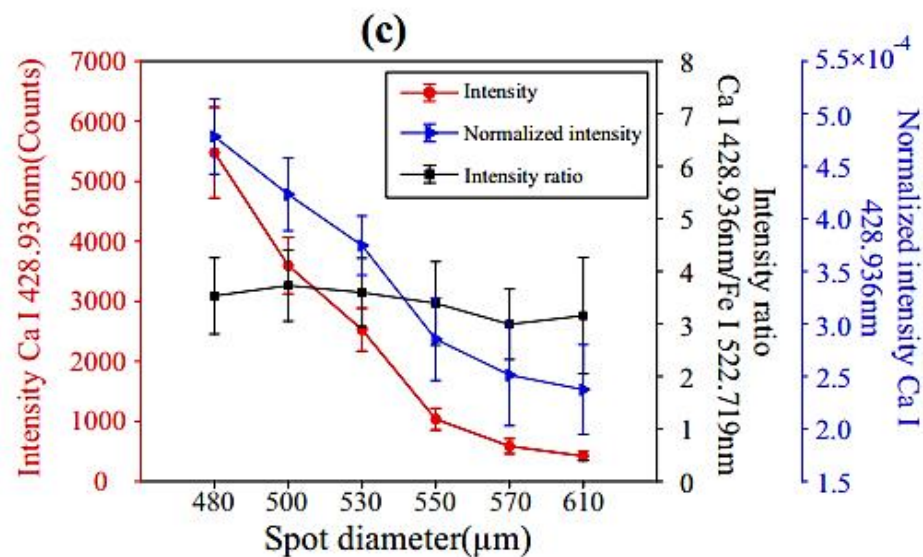
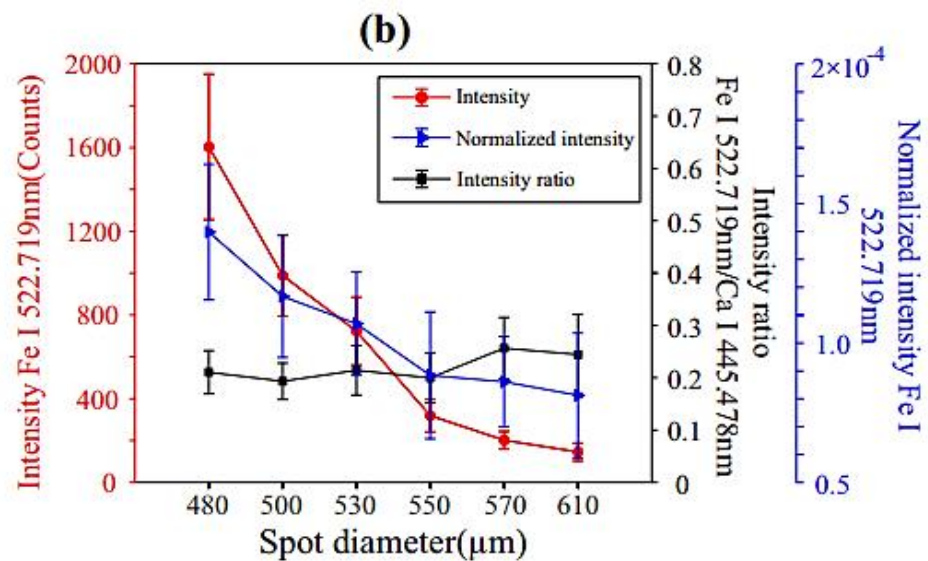
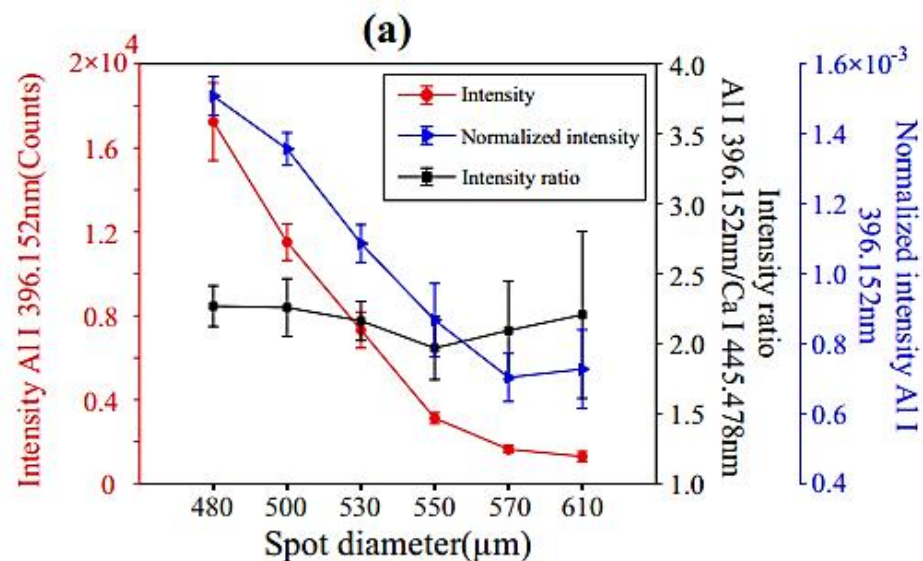


# 混淆矩阵图



```
clear;close all;clc
% 混淆矩阵
mat = [5,3,0;2,3,1;0,2,11];
% 标签
label = {'CAT','Dog','Rabbit'};
% 混淆矩阵主题颜色
% 可通过各种拾色器获得rgb色值
maxcolor = [191,54,12]; % 最大值颜色
mincolor = [255,255,255]; % 最小值颜色
% 绘制坐标轴
m = length(mat);
imagesc(1:m,1:m,mat)
xticks(1:m)
xlabel('Predict class','fontsize',10.5)
xticklabels(label)
yticks(1:m)
ylabel('Actual class','fontsize',10.5)
yticklabels(label)
% 构造渐变色
mymap = [linspace(mincolor(1)/255,maxcolor(1)/255,64)',...
linspace(mincolor(2)/255,maxcolor(2)/255,64)',...
linspace(mincolor(3)/255,maxcolor(3)/255,64)'];
colormap(mymap)
colorbar()
% 色块填充数字
for i = 1:m
    for j = 1:m
        text(i,j,num2str(mat(j,i)),...
'horizontalAlignment','center',...
'verticalAlignment','middle',...
'fontname','Times New Roman',...
'fontsize',10);
    end
end
% 图像坐标轴等宽
ax = gca;
ax.FontName = 'Times New Roman';
set(gca,'box','on','xlim',[0.5,m+0.5],'ylim',[0.5,m+0.5]);
axis square
% 保存
saveas(gca,'m.png');
```

# 带误差棒的折线图



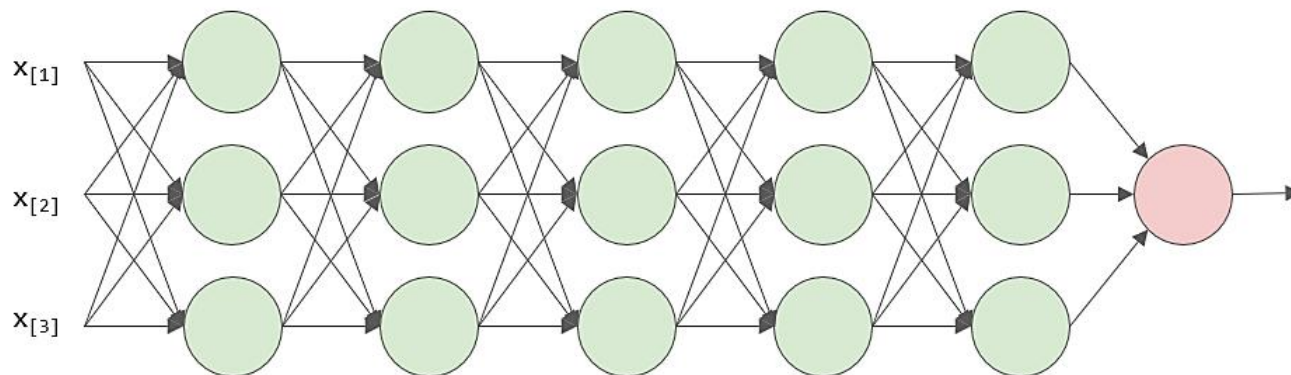
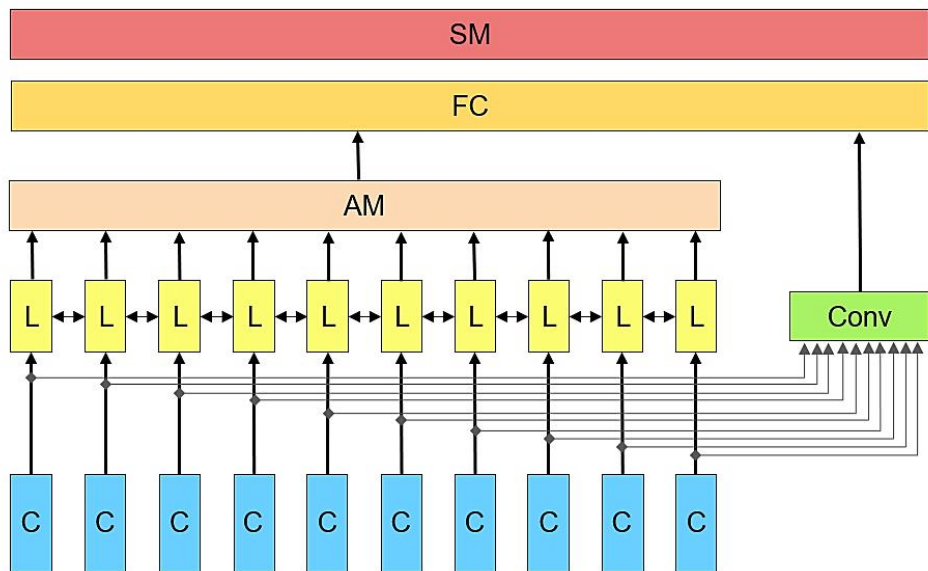
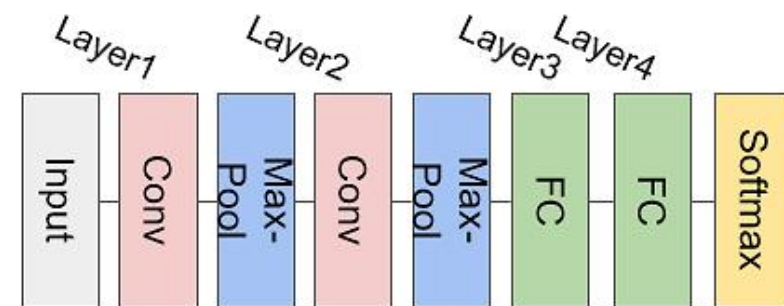
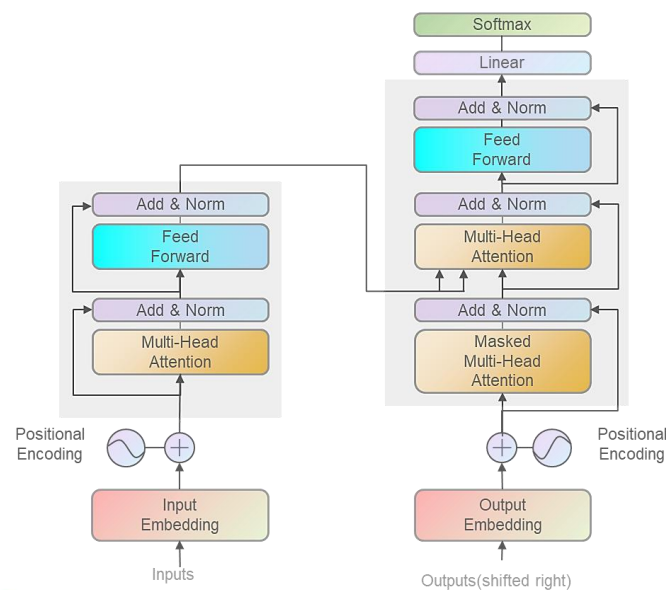
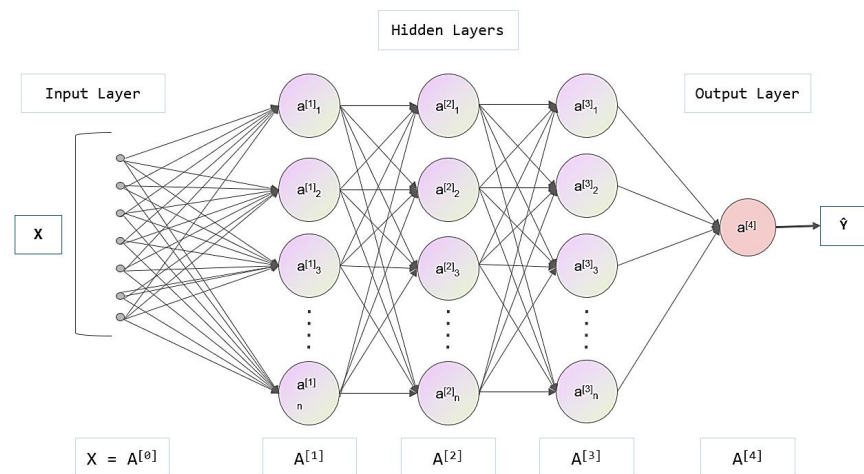




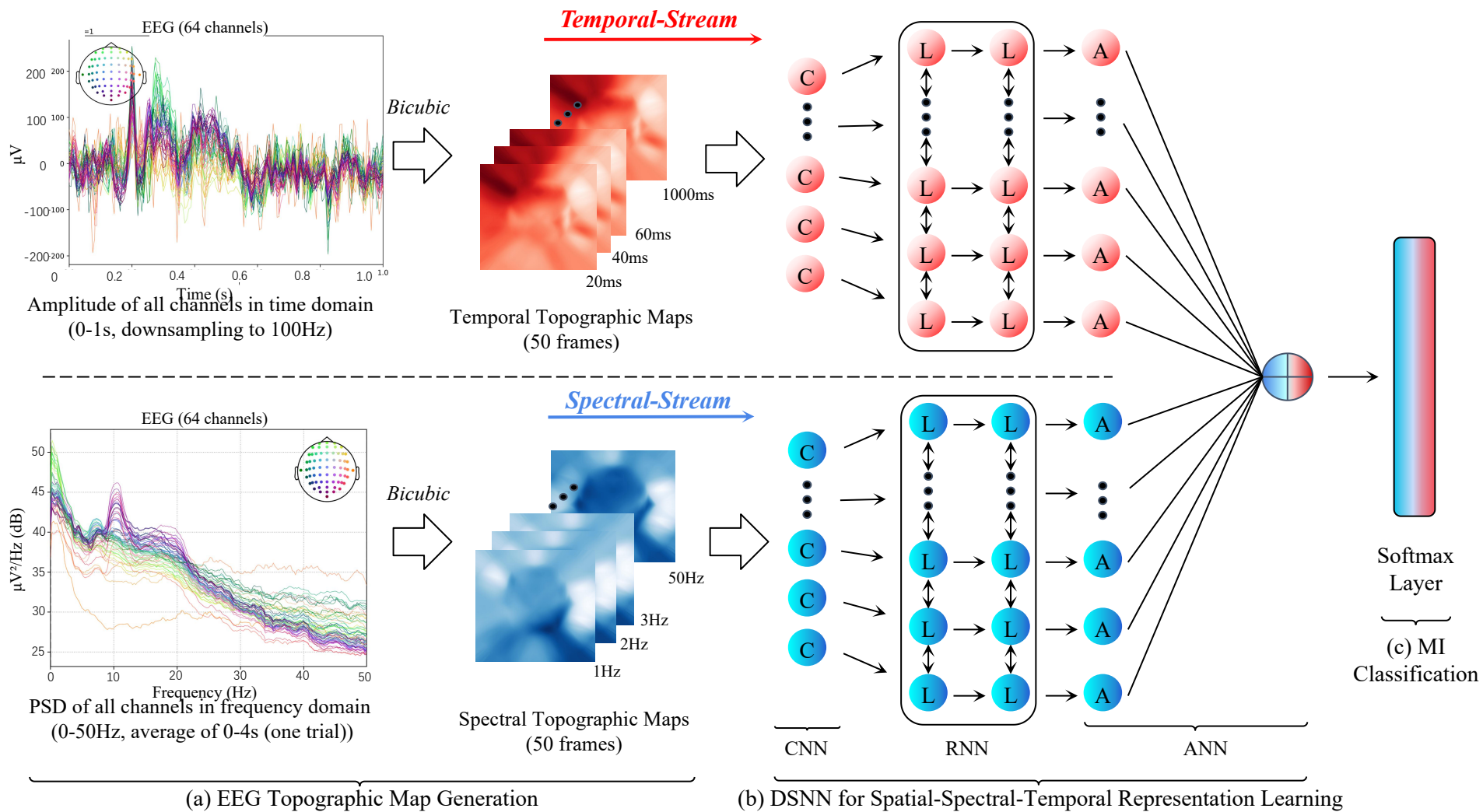
# 机器学习画图模板



# 常见机器学习作图组件



# EEG脑电识别网络结构实例





## 部分可视化工具及组件展示

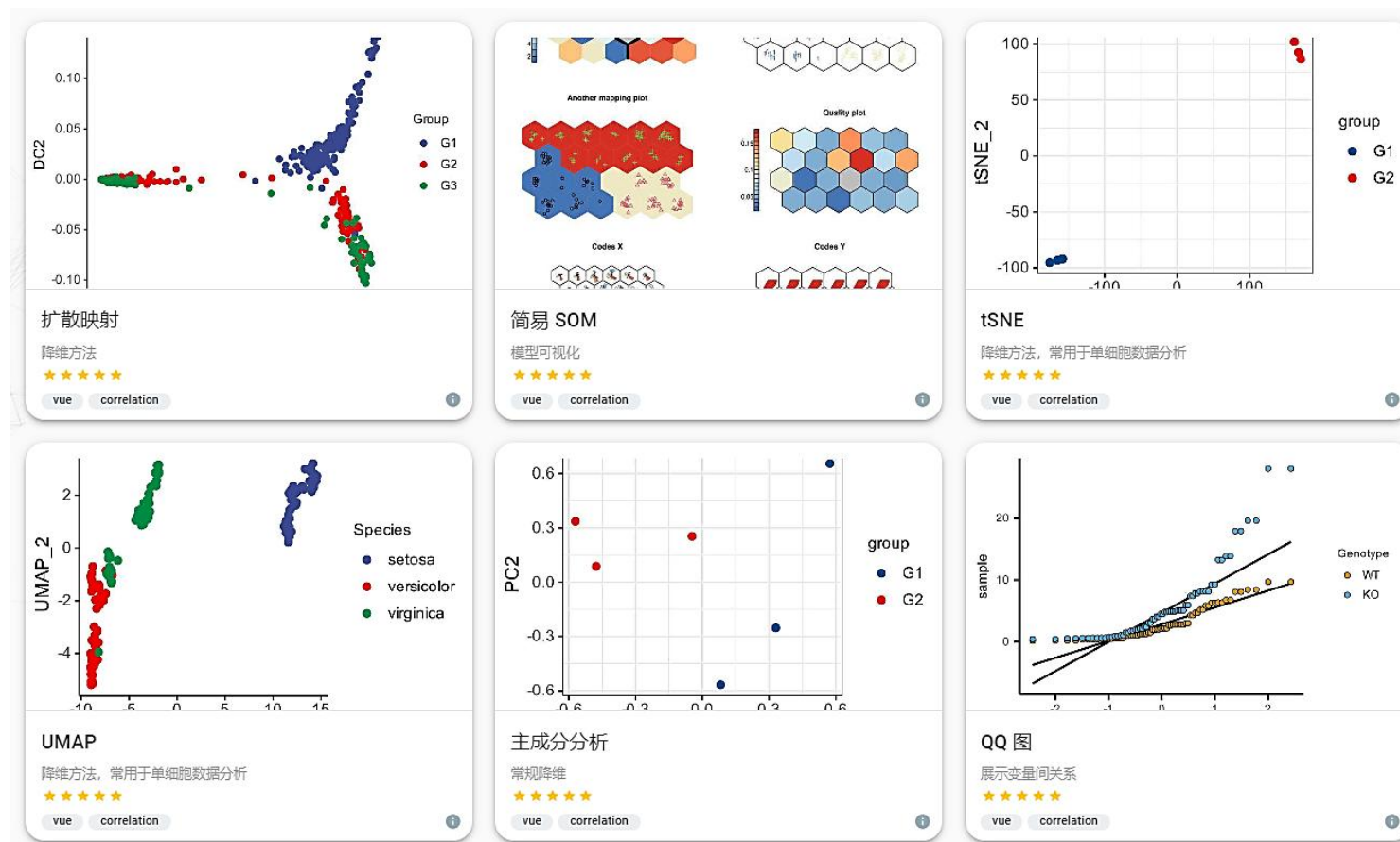




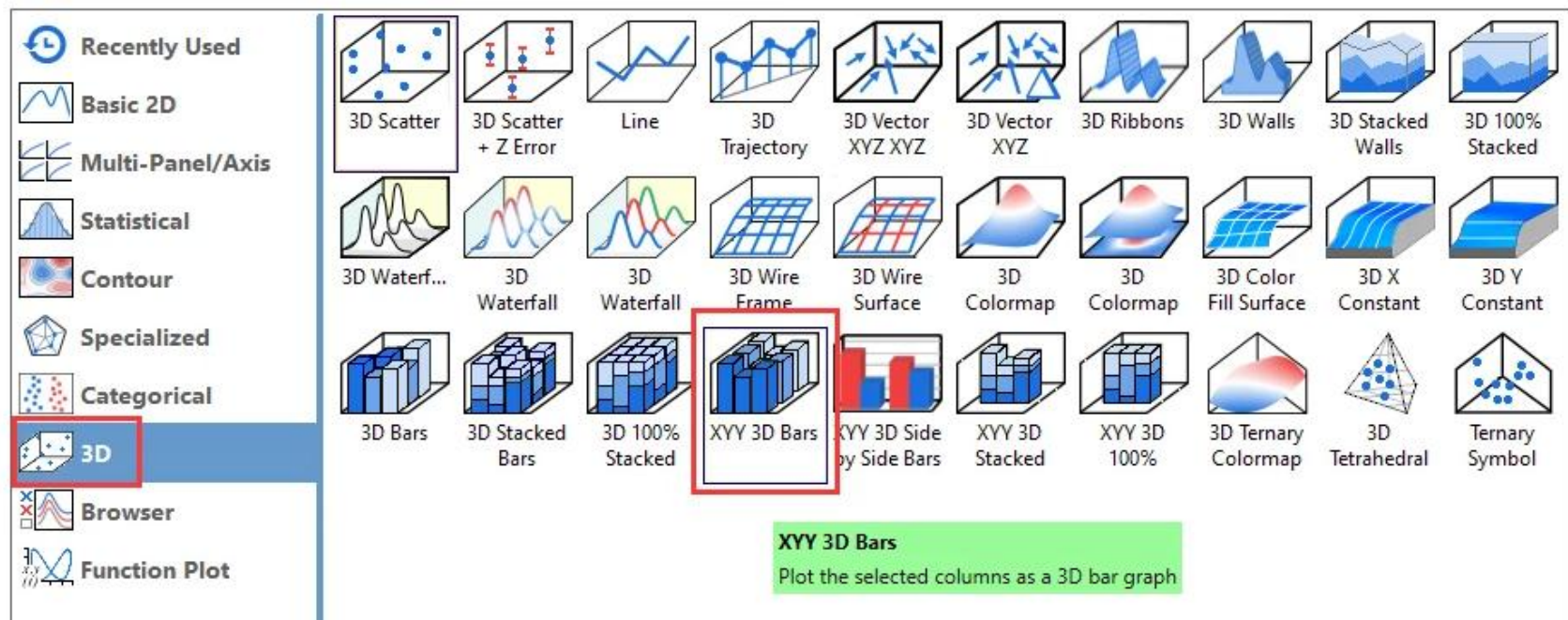
# 用于便捷可视化的网站——Hiplot生物医学数据分析

可以辅助缺少编程知识的生物学家/临床医生和经验丰富的生物信息学研究人员更好地利用他们的数据。目前，发表级别生物医学数据可视化分析在基于组学的平台中相对欠缺，而基于可视化的数据统计分析需求正在迅速增长。在此，我们开发了一个综合性生物医学**可视化分析工具箱**（240+）。其具有多样的功能任务插件，如基础统计、**回归**、**聚类分析**、基因组学、转录组学、**时间序列**、荟萃分析和风险模型等。与其他类似平台相比，它提供了更加易于使用的用户接口和交互方法。它由openbioX开源社区支持和运维。

可编辑的表格视图和切换的文件上传功能简化了大多数任务的数据加载过程。数据，参数和任务历史记录均可以通过 JSON 格式文件进行导入和导出，以增强任务的可重现性。单个 **R 语言脚本**或**函数**就可用于生成 Hiplot 工具插件的网页前端、命令行接口以及工具的描述信息。到目前为止，Hiplot 上已有数百个免费应用程序在提供服务，包括基本的统计描述，预测模型，多组学数据分析和文本挖掘等。

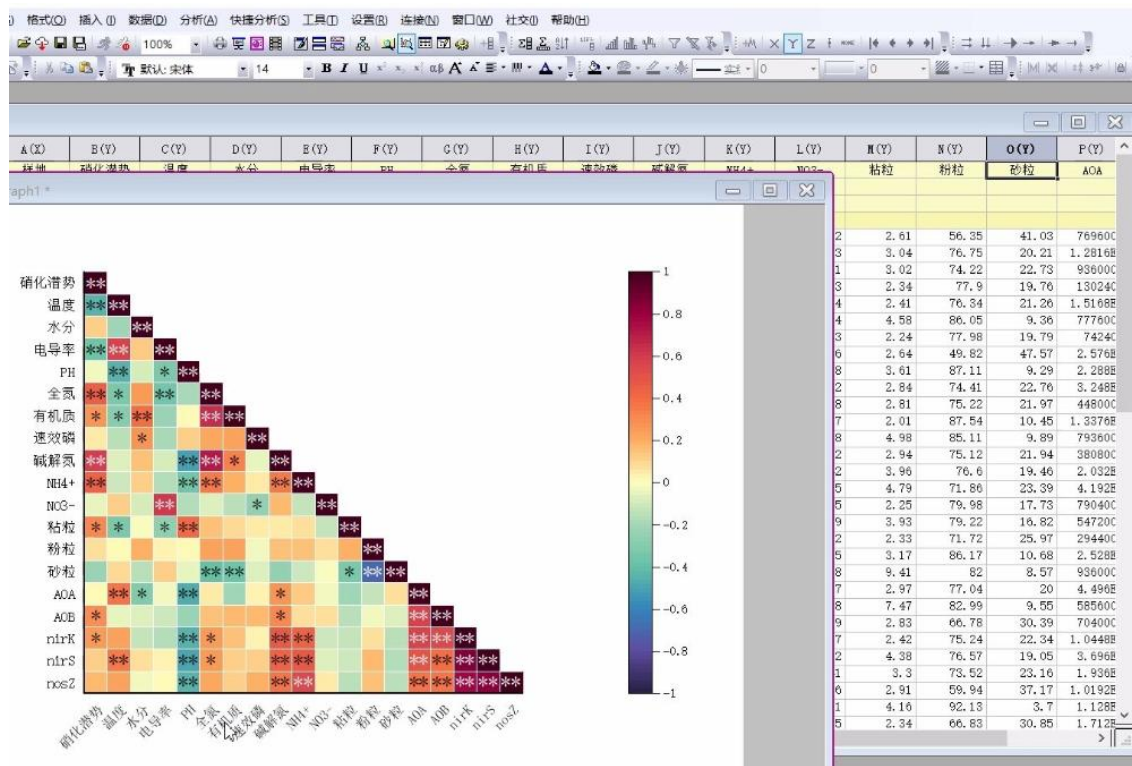


# Origin绘图工具

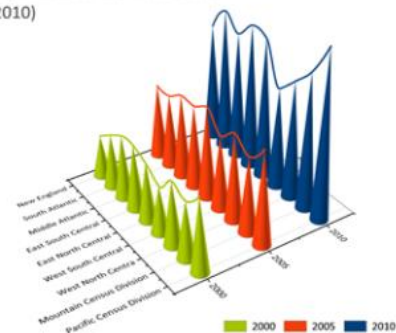


Origin是由OriginLab公司开发的一个科学绘图、数据分析软件，支持在Microsoft Windows下运行。Origin支持各种各样的2D/3D图形。Origin中的数据分析功能包括统计，信号处理，曲线拟合以及峰值分析。Origin中的曲线拟合是采用基于Levenberg-Marquardt算法（LMA）的非线性最小二乘法拟合。Origin强大的数据导入功能，支持多种格式的数据，包括ASCII、Excel、NI TDM、DIADem、NetCDF、SPC等等。图形输出格式多样，例如JPEG，GIF，EPS，TIFF等。内置的查询工具可通过ADO访问数据库数据。

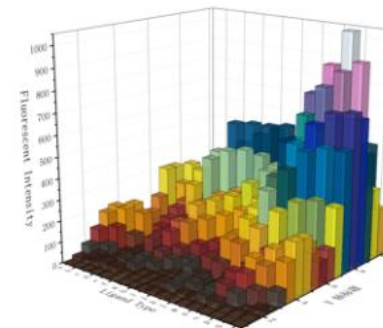
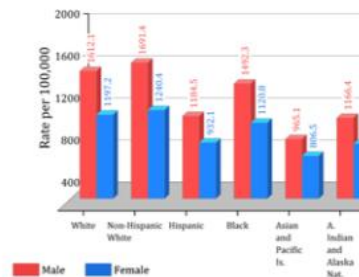
# Origin绘图工具



Employment Rate in United States  
(1 - 2010)



Cancer Incidence by Race/Ethnicity

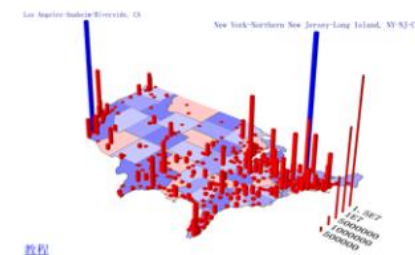


ORIGINPRO® 2022  
The Ultimate Software for Graphing & Analysis

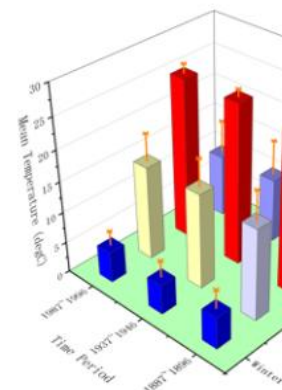
## 3D条状图



美国都市人口分布

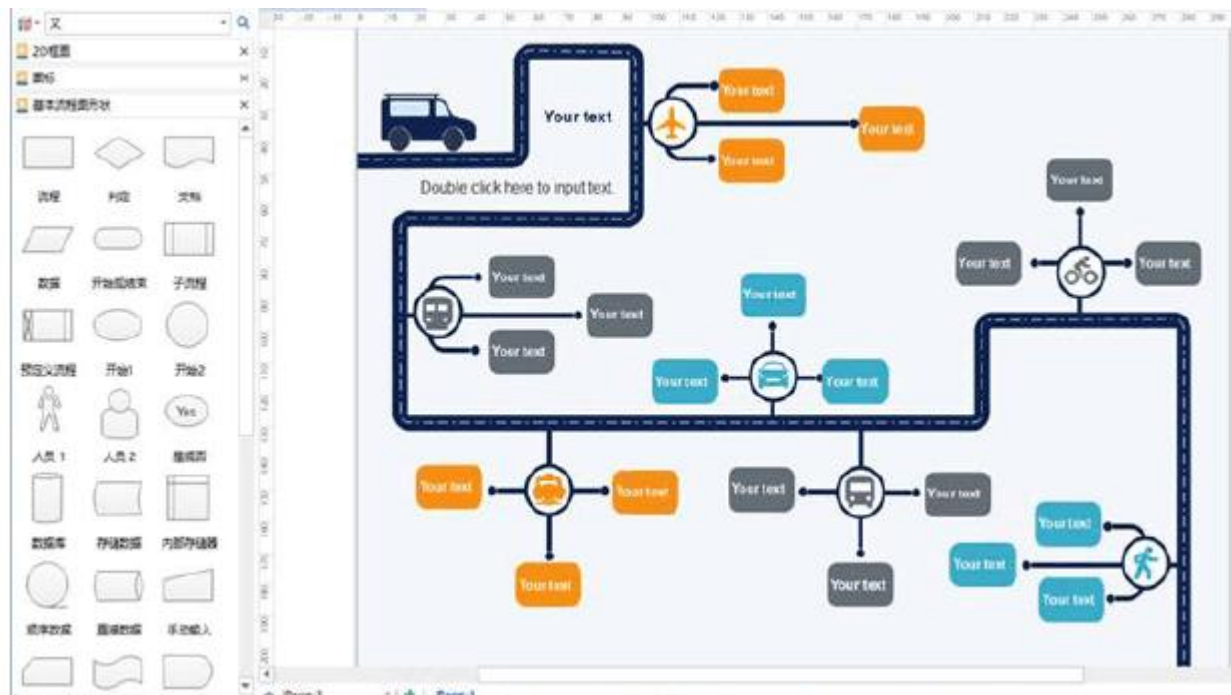


季度平均温度及标准差





# Visio绘图工具



Microsoft Visio是Windows操作系统下运行的流程图和矢量绘图软件，可以绘制流程图、UML用例图、宏观导图、类图(powerDesign)、层次方框图、warnier图、结构图、时序图、活动图、状态图、协作图、对象图、组件图、包图、部署图、思维导图等。



**thanks**

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